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ECA3102

Nano electronics

Exp6: MOSFET Scaling Analysis and Subthreshold Swing Calculation using

NanoHub

Aim: To analyze the effects of **MOSFET device scaling** and calculate the **subthreshold swing (SS)** using NanoHub simulations, in order to evaluate the switching performance and power efficiency of nanoscale MOSFETs.

Outcome:

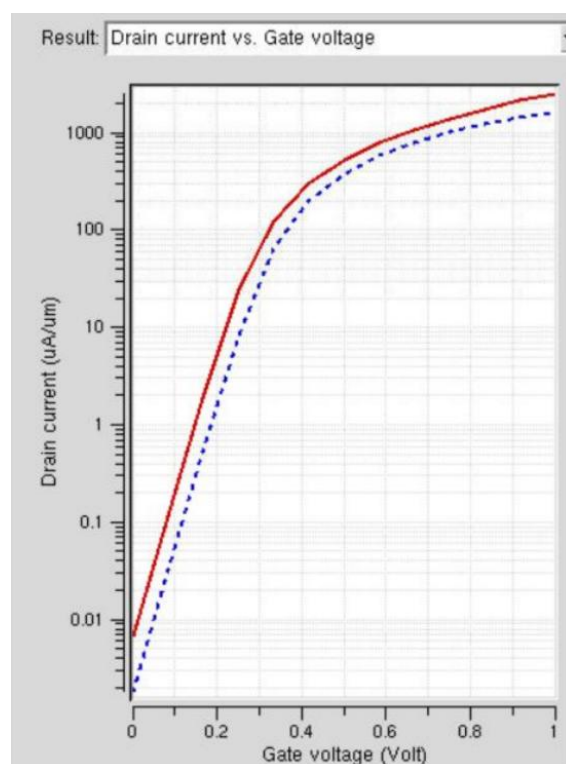
Successful simulation of MOSFETs with different scaling parameters.

Calculation of the **subthreshold swing (SS)** from the transfer characteristics.

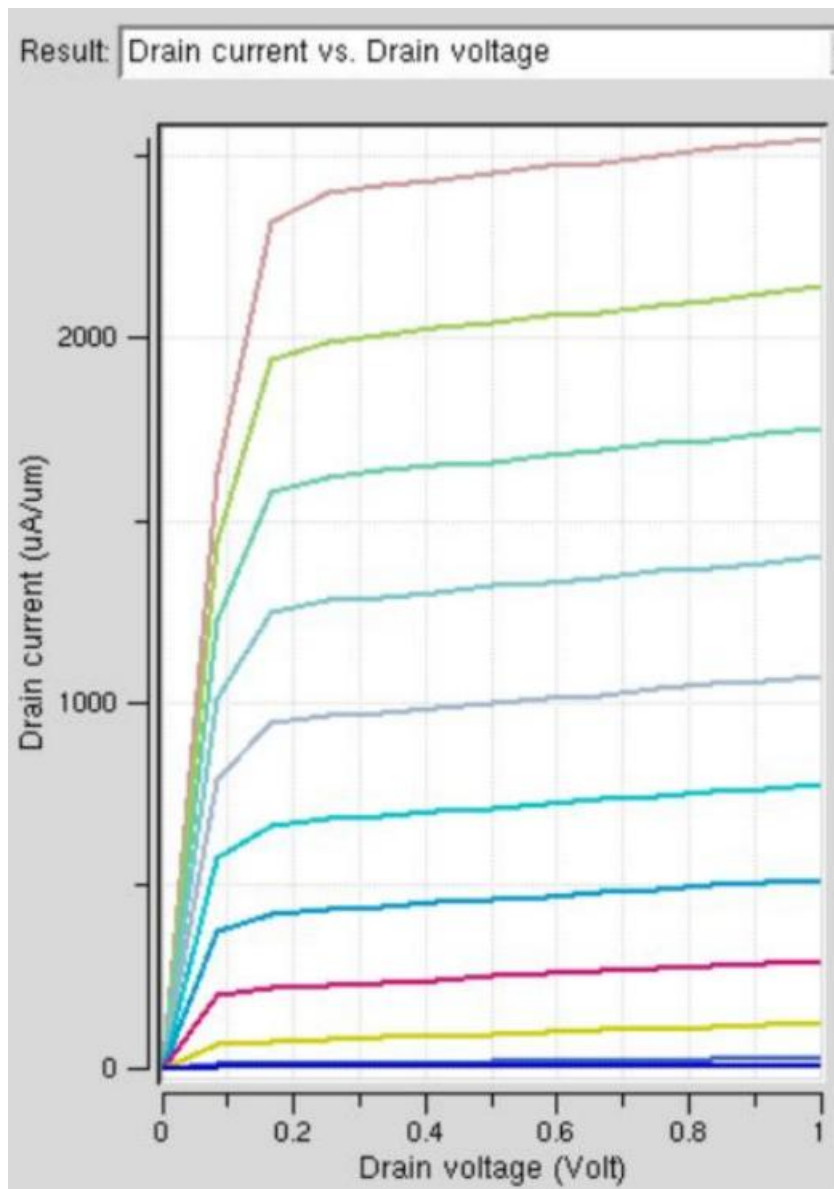
Analysis of the effect of scaling on threshold voltage, leakage current, and device performance.

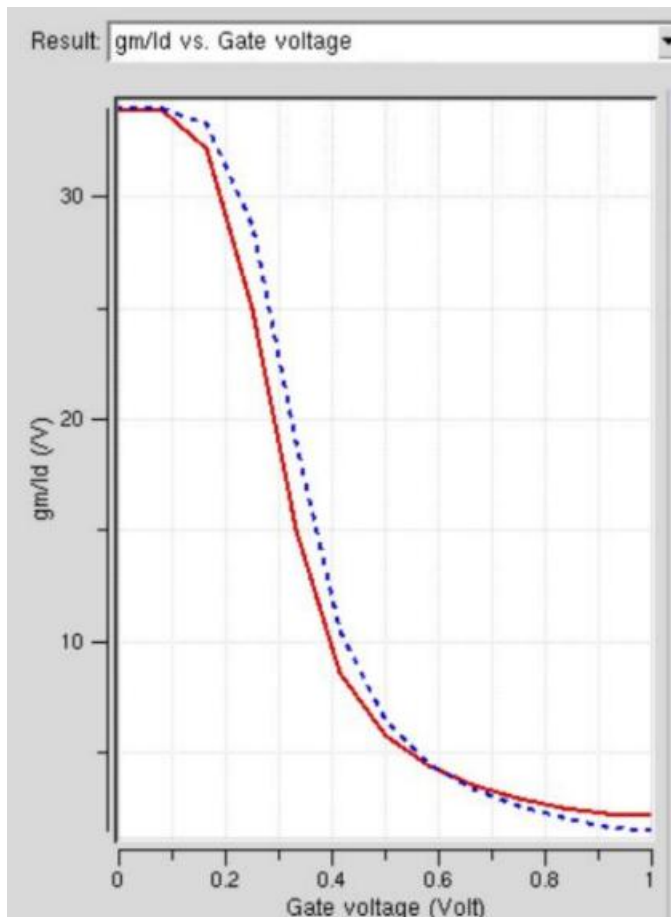
Better understanding of the trade-offs involved in designing **nanoscale CMOS devices** for modern VLSI, low-power electronics, and high-speed integrated circuits.

Case1: Case 1: Increase in V_G



Case 2: Variation in V_d :





Outcome justification: figure 1:

- At **low gate voltages**, the drain current is extremely small because the MOSFET is in the **OFF state**.
- Near the **threshold voltage (V_{TH})**, the drain current rises sharply, indicating channel formation.
- Beyond the threshold voltage, the MOSFET enters the **ON state**, where the drain current increases significantly.
- The **solid red curve** exhibits a slightly higher drain current than the **dashed blue curve**, indicating improved current drive and better switching performance.

Inference

The transfer characteristics confirm proper MOSFET operation and enable the calculation of **threshold voltage** and **subthreshold swing**, which are essential parameters for evaluating scaled MOSFET performance.

Figure 2:

- At **small drain voltages**, the drain current increases almost linearly with drain voltage, indicating the **linear (ohmic) region**.
- As the drain voltage increases further, the curves become nearly flat, indicating the **saturation region**.
- Each curve corresponds to a different gate voltage.
- Higher gate voltages produce higher drain currents because a stronger inversion channel is formed.
- The slight increase in current after saturation is due to **channel length modulation**, which is commonly observed in practical MOSFETs.

Inference

The output characteristics verify that the MOSFET successfully transitions from the linear region to the saturation region and demonstrate the influence of gate voltage on drain current, confirming proper device operation.

- Figure3: At **low gate voltages**, the g_m/I_D ratio is high, indicating excellent current efficiency in the **subthreshold region**.
- As the gate voltage increases, the ratio gradually decreases because the MOSFET moves into the **strong inversion region**.
- The solid and dashed curves follow similar trends, indicating consistent device behavior with slight differences due to scaling or device parameters.

Inference

The graph demonstrates the trade-off between **current efficiency** and **drive current**. High g_m/I_D values are desirable for low-power circuits, while lower values correspond to high-performance operation.